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FUTURE TECH SCHOOL PROGRAM

Faculty Name :

Course Duration : 5 Months

Classes : Offline & Online

Daily Teaching : 2 Hours

Total Classes : 100

Level 2 – Young Technologist (7th–8th Standard)

Lesson 22: For Loop in Python



Learning Objectives

By the end of this lesson, students will be able to:

- ✅ Understand what a **For Loop** is.
- ✅ Learn why loops are useful.
- ✅ Use the for loop with the range() function.
- ✅ Print numbers, patterns, and multiplication tables.
- ✅ Write simple programs using for loops.
- ✅ Solve real-life problems using repetition.

What is a Loop?

A **Loop** is a programming concept that **repeats a block of code multiple times**.

Instead of writing the same code again and again, we use a **loop**.

Real-Life Example

Imagine your teacher asks you to write:

"I will practice Python."

10 times.

❌ Without a loop, you would write the same sentence 10 times.

✅ With a **For Loop**, Python repeats it automatically.

Real-Life Example of a Loop



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Why Do We Use Loops?

Loops help us:

- 🔁 Repeat tasks automatically.
- ⏱️ Save time.
- 🖌️ Reduce code duplication.
- 📖 Make programs easier to read.
- 💻 Solve problems efficiently.

What is a For Loop?

A **For Loop** is used when we know **how many times** we want to repeat a task.

Syntax

```
for variable in range(number):
    statement
```

Example

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```
for i in range(5):
    print("Hello")
```

Output

Hello

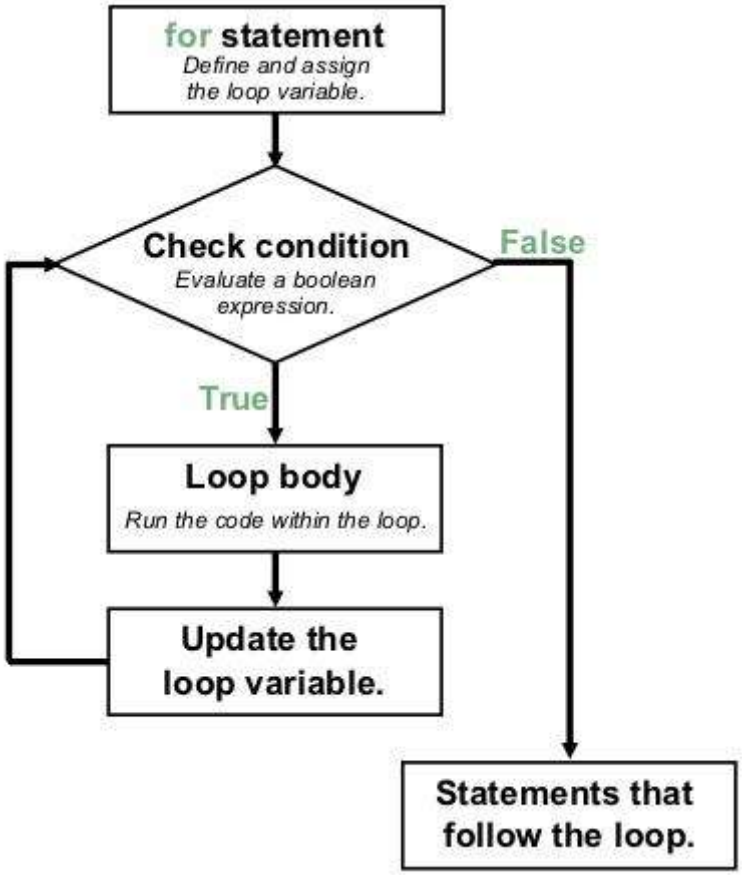
Hello

Hello

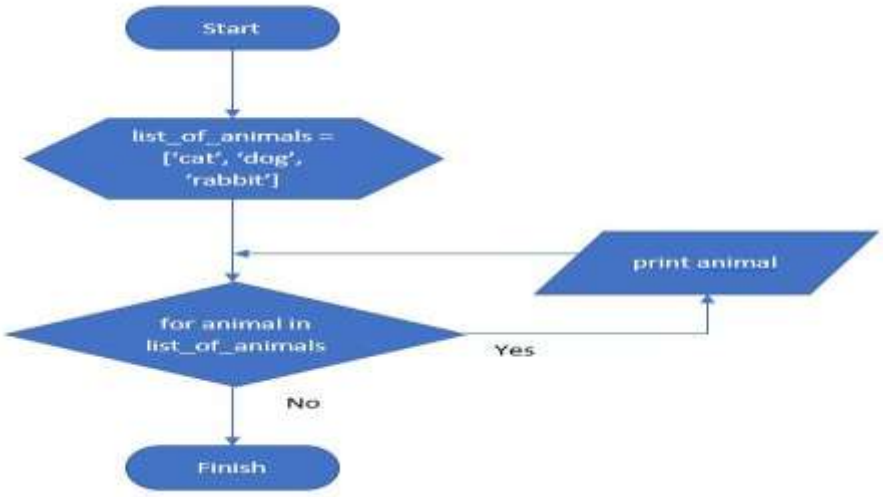
Hello

Hello

How a For Loop Works







Understanding range()

The **range()** function generates a sequence of numbers.

Example

```
for i in range(5):
    print(i)
```

Output

0

1

2

3

4

By default, range(5) starts at **0** and stops **before 5**.

Types of range()

Syntax	Meaning	Example Output
range(5)	0 to 4	0,1,2,3,4
range(1,6)	1 to 5	1,2,3,4,5
range(2,11,2)	Even numbers	2,4,6,8,10

Example 1

Print numbers from **1 to 5**.

```
for i in range(1, 6):
    print(i)
```

Output

1

2

3

4

5

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Example 2

Print even numbers.

```
for i in range(2, 11, 2):
   print(i)
```

Output

2

4

6

8

10

Example 3

Print odd numbers.

```
for i in range(1, 10, 2):
   print(i)
```

Output

1

3

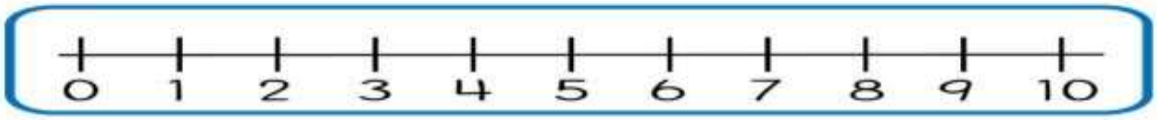
5

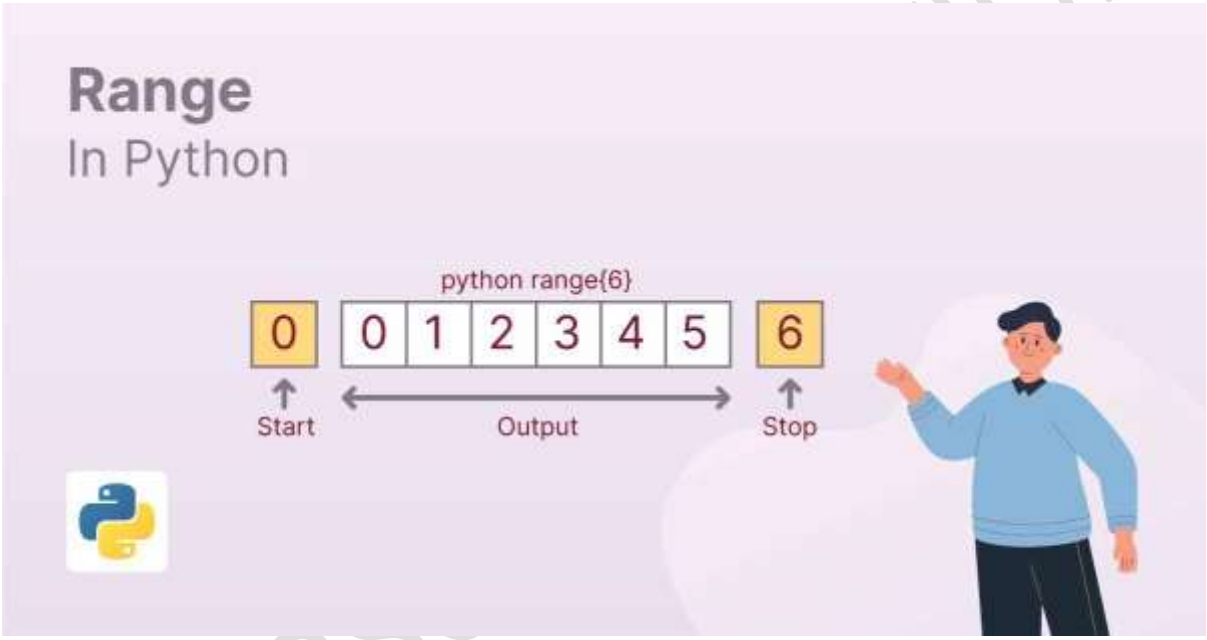
7

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Understanding range()

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Printing a Name Multiple Times

```
for i in range(5):
    print("Rahul")
```

Output

Rahul

Rahul

Rahul

Rahul

Rahul

Multiplication Table of 5

```
for i in range(1, 11):
    print("5 x", i, "=", 5 * i)
```

Output

```
5 x 1 = 5
5 x 2 = 10
...
5 x 10 = 50
```

Sum of Numbers from 1 to 5

```
total = 0

for i in range(1, 6):
    total = total + i

print("Total =", total)
```

Output

```
Total = 15
```

Printing a Star Pattern

```
for i in range(5):
    print("*")
```

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Output

*

*

*

*

*

Square Numbers

```
for i in range(1, 6):
   print(i ** 2)
```

Output

1

4

9

16

25

Using a For Loop with a List

```
fruits = ["Apple", "Banana", "Mango"]

for fruit in fruits:
   print(fruit)
```

Output

Apple

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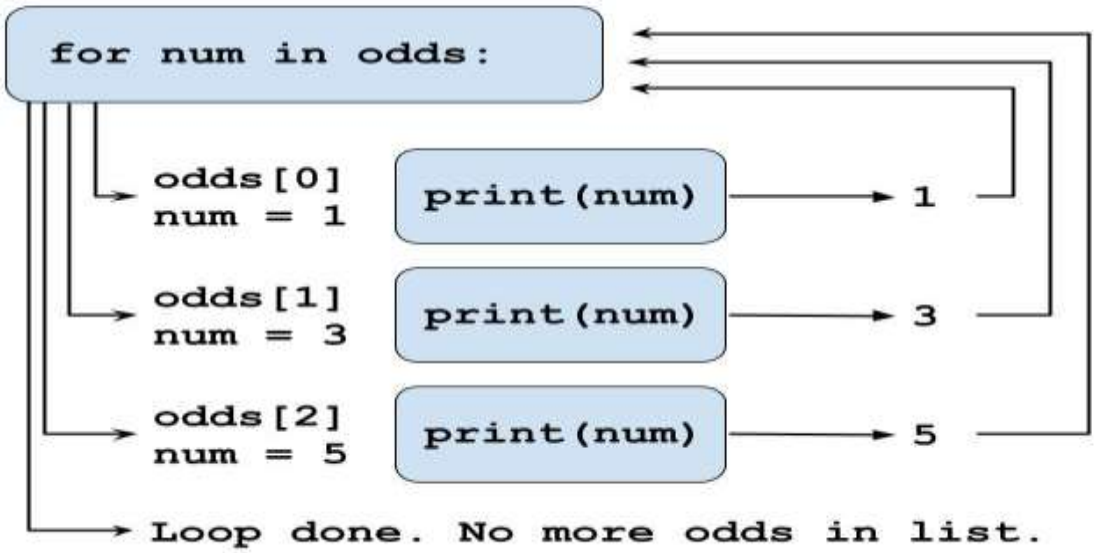
Banana

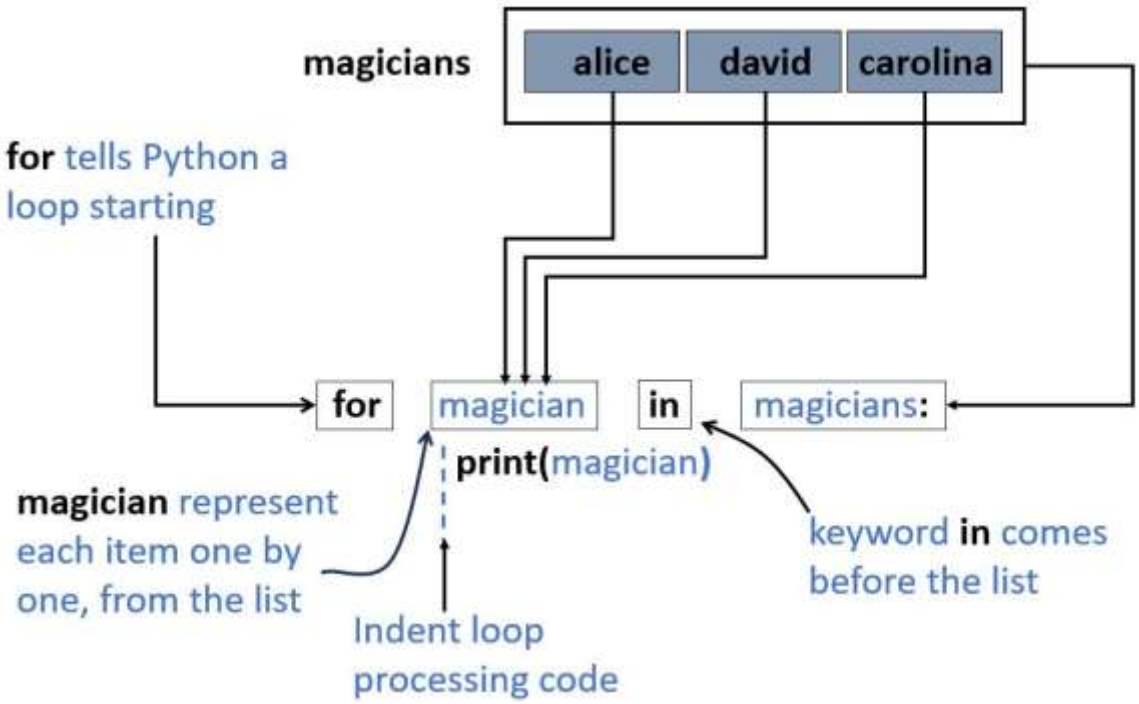
Mango

Loop Through a List




```
odds = [1, 3, 5]
for num in odds:
    print(num)
```





Practical Session

Practical Activity 1

Print Numbers 1 to 10

```
for i in range(1, 11):
    print(i)
```

Practical Activity 2

Print Your Name 10 Times

```
name = input("Enter Your Name: ")


for i in range(10):
    print(name)
```

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Practical Activity 3

Multiplication Table

```
number = int(input("Enter a Number: "))

for i in range(1, 11):
    print(number, "x", i, "=", number * i)
```

Practical Activity 4

Print Even Numbers

```
for i in range(2, 21, 2):
    print(i)
```

Practical Activity 5

Print Odd Numbers

```
for i in range(1, 20, 2):
    print(i)
```

Practical Activity 6

Calculate Sum of 1 to 10

```
total = 0

for i in range(1, 11):
    total += i
```

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```
print("Sum =", total)
```

Practical Activity 7

Print Squares

```
for i in range(1, 11):
    print(i, "Square =", i ** 2)
```

Practical Activity 8

Display Student Subjects

```
subjects = ["Math", "Science", "English", "Computer"]

for subject in subjects:
    print(subject)
```

Classroom Coding Activity



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Activity 1

Print numbers from **1 to 20**.

Activity 2

Print the multiplication table of **8**.

Activity 3

Print the squares of numbers from **1 to 10**.

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Activity 4

Create a list of your favorite fruits and print each fruit using a for loop.

Real-Life Applications

Situation	Use of For Loop
Attendance List	Print student names
Multiplication Tables	Repeat calculations
Drawing Shapes	Print patterns
Games	Repeat game actions
Calendar	Display days or months
Scoreboard	Show scores one by one

Common Mistakes

❌ **Missing Colon (:)**

Wrong

```
for i in range(5)
   print(i)
```

✅ **Correct**

```
for i in range(5):
   print(i)
```

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❌ **Incorrect Indentation**

Wrong

```
for i in range(5):
print(i)
```

✅ **Correct**

```
for i in range(5):
    print(i)
```

❌ **Wrong range()**

Wrong

```
range(1,5)
```

This prints:

1

2

3

4

It **does not include 5**.

Best Practices

- ✓ Use meaningful variable names like number, fruit, or student.
- ✓ Indent the code inside the loop.
- ✓ Use range() carefully (remember the ending number is not included).

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- ✓ Test your program with different values.
- ✓ Add comments to explain your code.

Keywords

- **Loop** – Repeats a block of code.
- **For Loop** – Repeats code a fixed number of times.
- **range()** – Generates a sequence of numbers.
- **Iteration** – One repetition of a loop.
- **Indentation** – Spaces that define the loop block.

Lesson Summary

- A **For Loop** repeats code automatically.
- It is useful when the number of repetitions is known.
- The range() function generates sequences of numbers.
- for loops can work with numbers, strings, and lists.
- Loops make programs shorter, cleaner, and more efficient.

Quiz

Multiple Choice Questions

1. Which keyword is used to create a loop in Python?

A) if

B) for

C) print

D) input

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✅ **Answer: B**

2. What is the output of range(5)?

A) 1, 2, 3, 4, 5

B) 0, 1, 2, 3, 4

C) 0, 1, 2, 3, 4, 5

D) 5

✅ **Answer: B**

3. Which function is commonly used with a for loop?

A) print()

B) input()

C) range()

D) len()

✅ **Answer: C**

4. What will this code print?

```
for i in range(3):
    print("Hi")
```

A) Hi (1 time)

B) Hi (2 times)

C) Hi (3 times)

D) Error

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✅ **Answer: C**

5. Why do we use a for loop?

A) To compare values

B) To repeat tasks

C) To take input

D) To create variables

✅ **Answer: B**

Homework

1. Write a program to print numbers from **1 to 50**.
2. Write a program to print even numbers from **2 to 20**.
3. Write a program to print odd numbers from **1 to 19**.
4. Write a program to display the multiplication table of **9**.
5. Write a program to calculate the sum of numbers from **1 to 20**.
6. Create a list of **5 animals** and print each one using a for loop.
7. Print the cubes of numbers from **1 to 10**.

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